

Span2Type: Open World Named Entity Recognition with Automatic Entity Naming

Master's Thesis Defense

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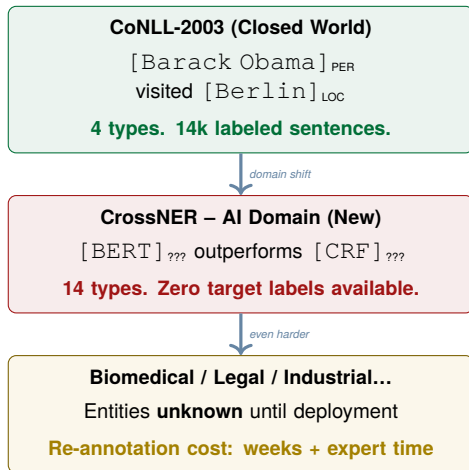
Chang Gung University, Taiwan

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What NER does (closed benchmarks)

- Locate entity spans in text
- Classify into *fixed, predefined* types:
PERSON · ORG · LOCATION
- BERT/RoBERTa: near-human F1 on CoNLL-2003



Three Concrete Deployment Challenges

#1 Domain Shift

BIO tagger trained on newswire fails on AI / bio / legal text.

Multi-token technical entities:
support vector machine
recurrent neural network

Single-view BIO misses boundaries.

#2 Unknown Type Inventory

New domains have:

- Finer-grained types
- Entirely new categories
- Evolving terminology

Supervised models cannot discover *new* types.

Need: automatic k selection.

#3 Interpretability Gap

Unsupervised clustering produces:

Cluster 3, Cluster 7, ...

Anonymous IDs → expert must manually inspect & label.

Defeats the purpose of unsupervised discovery.

Method	Ch. 1 Span Detection	Ch. 2 Auto-k Typing	Ch. 3 Naming
Supervised NER	×	×	×
Few-shot NER	◇	×	×
OWNER (baseline)	◇	✓	×
Span2Type (ours)	✓	✓	✓

Common limitation across all eras:

- Require predefined type inventory
- Need at least a few labeled examples per type
- Cannot discover *new* entity categories

Open-world NER removes all constraints:

- No predefined types
- No target-domain annotations
- Discovers types automatically via clustering

Sequence tagging (BIO):

- Standard: BERT/DeBERTa + single BIO head
- **Error propagation**: wrong B tag corrupts entire span
- Single boundary signal — easily confused cross-domain

Span enumeration (PURE, Chen '21):

- Enumerate all $\mathcal{O}(n^2)$ candidate spans explicitly
- No boundary cascading errors
- **Weak cross-domain transfer** — F1 35.8 avg on CrossNER

Prototype / coref-based detection:

- **SpanProto**: prototype matching — F1 61.0 avg
- **WL-Coref**: coreference boundaries — F1 65.6 avg
- Still limited to a *single* detection view

Multi-view ensemble (DTrans, Zheng '22):

- Three complementary decoders trained jointly:
BIO + Start-End (SE) + Tie-Break (TB)
- Mean Teacher EMA enforces cross-view consistency
- Errors in one view cancelled by consensus of others
- **F1 82.4 avg — strongest cross-domain detection**

Why single-view is insufficient for open-world NER

Cross-domain entities have diverse surface forms. A single boundary representation overfits source-domain patterns; multi-view consensus is more robust.

Span2Type is the *first* open-world NER system to integrate DTrans multi-view detection.

Entity type list generation:

- **Supervised NER**: human-crafted fixed inventory (PER, ORG, LOC...) — **requires domain experts**
- **Few-shot NER** (FewNERD, EpiMeta): type names still **required at test time** from the user
- **OWNER '23**: auto-discover types via clustering — first system to generate type list automatically
- **Limitation**: cluster names still coarse/redundant

Parameter-efficient fine-tuning:

- Full fine-tuning: 110M+ parameters updated
- Adapters (Houlsby '19): bottleneck layers, $\sim 3\text{M}$ params
- **Soft prompts** (Lester '21): k trainable prefix tokens, PLM weights fully frozen — **<0.004% of params**
- LoRA (Hu '22): low-rank weight decomposition

Cluster naming — exemplar selection strategies:

- **OWNER (MLM)**: uses *all* entities in a cluster — no sampling, but **noisy & slow on large clusters**
- **OWNER (LLM) / Random**: subsample $n=16$ randomly — cheap, **unstable** (misses semantic range)
- **Centroid-nearest** (Span2Type ablation): representative anchor, but members are **mutually redundant**
- **MMR** (Carbonell '98): Maximal Marginal Relevance
 - ▶ Jointly optimises relevance *and* diversity
 - ▶ Each new exemplar maximises marginal gain

Three open problems in open-world NER

- 1 No multi-view span detection adopted

OWNER pipeline — 3 stages:

1 Mention Detection

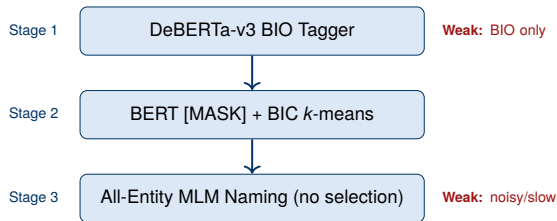
DeBERTa-v3-base + *single* BIO head
Trained on source domain

2 Entity Typing

BERT fill-in-the-blank template
k-means with BIC-guided auto-*k*

3 Cluster Naming

All cluster entities — no selection
Single template + BERT MLM majority-vote



Strong baseline: 49.4 avg AMI on CrossNER

Gap 1 — Span Detection

No open-world NER system uses **multi-view** span detection for cross-domain robustness.

Gap 2 — Entity Typing Efficiency

Soft prompt tuning has not been applied to entity type embedding learning in open-world NER.

Gap 3 — Exemplar Diversity

Centroid-nearest / random exemplar selection for cluster naming is sub-optimal and unstable.

Overarching Contribution

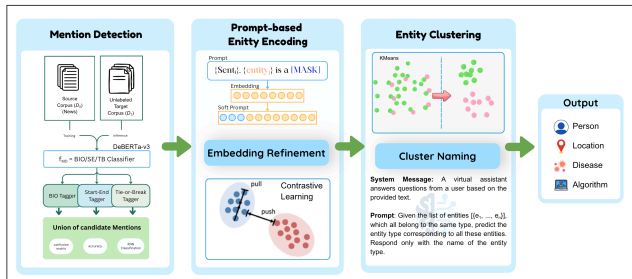
Build a complete framework: given an **unannotated target corpus**, automatically produce entity spans each assigned a **human-readable type name** — no predefined type list, no target-domain labels.

Technical Objectives:

- 1 Integrate **DTrans multi-view** detection (BIO + SE + TB) into the OWNER pipeline
- 2 Develop **two-stage soft-prompt** entity typing — better initialization, minimal forgetting
- 3 Propose **MMR-based** exemplar selection for more specific, stable cluster naming

Key constraint

Zero target-domain supervision at any stage.



Three-stage pipeline:

- 1 Entity Detection**
DTrans: BIO + Start-End + Tie-Break decoders
replaces OWNER's BIO-only tagger
- 2 Entity Typing**
BERT [MASK] template
+ soft prompt variant
+ BIC k -means clustering
- 3 Cluster Naming**
MMR exemplar selection
+ MLM or LLM backend
replaces all-entity / random

Key property

No target-domain labels at any stage.

Mapping Challenges to Contributions

Framework contribution

Span2Type: unannotated corpus $\xrightarrow{\text{zero supervision}}$ entity spans + human-readable type list (automatic entity type discovery)

Challenge	Root Cause	Span2Type Solution
#1 Robust span detection	BIO sees only one view of entity boundaries	DTrans: BIO + Start-End + Tie-Break, jointly with Mean Teaching
#2 Unknown type inventory	Fixed label space in supervised models	BIC-guided <i>k</i>-means — retained from OWNER
#3 Interpretability gap	Cluster IDs carry no semantic content	MMR exemplar selection + LLM naming backend
<i>Cross-cutting: typing quality</i>		Two-stage soft prompt: frozen warmup → full fine-tuning at low LR

Pipeline cascade effect

Better spans → cleaner embeddings →
more coherent clusters → better names

Improvements compound:

OWNER: 49.4 avg AMI

Span2Type: 61.4 avg AMI

$\Delta = +12.0$ points

Why single-view BIO is insufficient:

- Classifies each token as B/I/O in isolation
- Struggles with multi-token technical spans:
support vector machine
convolutional neural network
- One wrong token-tag → entire span lost

Three complementary views of a span:

- 1 **BIO** — global sequential context
- 2 **Start-End (SE)** — explicit boundary positions
- 3 **Tie-Break (TB)** — pairwise token adjacency

Key insight

Union of three views → **higher recall**
under domain shift

Example: “support vector machine”

tokens	support	vector	machine	achieved
BIO	B	I	I	O
SE start	1	0	0	0
SE end	0	0	1	0
TB	Tie	Tie	Break	All three

views agree on the same 3-token span.

Even if BIO mislabels one token, SE or TB recovers it.

Backbone: BERT-base-cased (12L, hidden $d = 768$)

Input $\mathbf{x} \rightarrow \text{encoder} \rightarrow \mathbf{H} \in \mathbb{R}^{T \times d}$

Three parallel heads:

BIO decoder:

$$p_t^{\text{BIO}} = \text{softmax}(\mathbf{W}_{\text{bio}} \mathbf{h}_t)$$

where $\mathbf{W}_{\text{bio}} \in \mathbb{R}^{3 \times d}$

Start-End decoder:

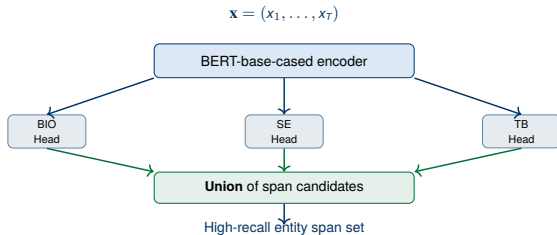
$$p_t^{\text{end}} = \text{softmax}(\mathbf{W}_{\text{end}}[\mathbf{h}_t; \hat{p}_t^{\text{start}}])$$

End conditions on predicted start $\Rightarrow$ boundary coherence

Tie-Break decoder:

$$p_t^{\text{tb}} = \text{softmax}(\mathbf{W}_{\text{tb}}(\mathbf{h}_t + \mathbf{h}_{t+1}))$$

Total loss:



Mean Teaching (EMA)

$$\tilde{\theta}_t = \alpha \tilde{\theta}_{t-1} + (1 - \alpha) \theta_t$$

$\alpha = 0.995$; soft pseudo-labels $\rightarrow$

auxiliary consistency loss

Template construction:

$$\underbrace{[\text{CLS}] x_1 \cdots x_T}_{\text{sentence}} \underbrace{m}_{\text{entity}} \text{ is a } \underbrace{[\text{MASK}]}_{\text{type}} . [\text{SEP}]$$

- Repeating the span m gives BERT an explicit type cue
- Leverages BERT's cloze pretraining knowledge

Entity embedding:

$$\mathbf{e}_s = \text{Dropout}(\mathbf{h}_{[\text{MASK}]}) \in \mathbb{R}^d$$

Training: Batch Triplet Margin Loss

$$\mathcal{L} = \frac{1}{|\mathcal{T}|} \sum_{(i,j,k) \in \mathcal{T}} \max(0, d_{ij} - d_{ik} + \gamma)$$

Concrete Example

Sentence: *Albert Einstein was born in Germany*

Span: *Albert Einstein*

Template:

[CLS] Albert Einstein was born in Germany **Albert Einstein** is a [MASK] . [SEP]

BERT predicts: *physicist, scientist*

$\mathbf{e}_s$ = [MASK] hidden state

→ rich type-semantic representation

After triplet training



Embedding space (t-SNE)

Full fine-tuning problem:

- Updates all $\sim 110\text{M}$ BERT parameters
- Risk of catastrophic forgetting of pretrained world knowledge
- Cross-domain: frozen backbone is safer

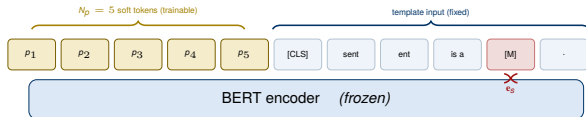
Soft prompt (P-tuning / Prefix-tuning):

- $N_p = 5$ trainable tokens $\mathbf{P} \in \mathbb{R}^{N_p \times d}$
- Prepended to template embeddings:

$$\text{inputs} = \left[\underbrace{\mathbf{P}}_{\text{trainable}}; \underbrace{\mathbf{E}_x}_{\text{frozen}} \right] \in \mathbb{R}^{(N_p+L) \times d}$$

- BERT backbone: **strictly frozen**
- Trainable params: $5 \times 768 = \mathbf{3,840}$

MASK index shifted by N_p



Parameter comparison

Full BERT fine-tuning	$\sim 110,000,000$
Soft prompt only	3,840
Reduction	$\times 28,645$

Two-Stage Prompt-Initialized Fine-Tuning

Problem:

- Soft-only $\rightarrow$ limited expressiveness
- Full fine-tuning from epoch 1 $\rightarrow$ catastrophic forgetting

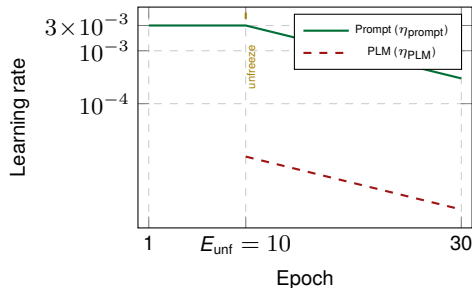
Solution: two-stage schedule at $E_{\text{unfreeze}} = 10$

	Stage 1	Stage 2
	Epochs 1–9	Epochs 10–30
Trainable	prompt only	prompt + BERT
LR (prompt)	3×10^{-3}	3×10^{-3}
LR (BERT)	frozen	1×10^{-5}
PLM frozen	✓	✗

Critical implementation details:

- `add_param_group()` — preserves optimizer moments for soft tokens

Learning Rate Schedule



Why this works

Stage 1: prompt warms up rapidly (3×10^{-3}) without disturbing pretrained BERT.

Stage 2: BERT adapts at 1×10^{-5} — $300\times$ smaller — preserving pretrained representations.

Challenge: target domain has *unknown* k

AutoKmeans: search $k \in [2, 30]$, step 2

Bayesian Information Criterion:

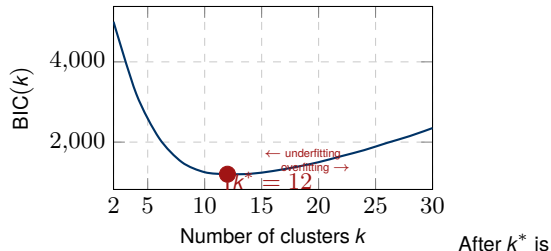
$$\text{BIC}(k) = n \cdot \log\left(\frac{\mathcal{I}(k)}{n}\right) + \log(n) \cdot k$$

- $\mathcal{I}(k)$: inertia (sum of squared distances)
- First term: fit quality — decreases with k
- Second term: complexity penalty — grows with k

$$k^* = \arg \min_k \text{BIC}(k)$$

- k-means++ init, 10 random restarts per k
- **Retained unchanged from OWNER**

BIC curve (schematic)



selected, embeddings $\{\mathbf{e}_s\}$ are assigned cluster labels $c_s \in \{1, \dots, k^*\}$ for all target-domain spans.

Problem with centroid-nearest selection:

- 16 exemplars all near centroid → redundant
- Single surface form repeated → unstable naming

Maximal Marginal Relevance (MMR):

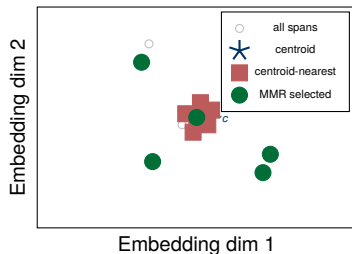
$$\text{MMR}(s) = \lambda \cdot \cos(\mathbf{e}_s, \boldsymbol{\mu}_c) - (1 - \lambda) \cdot \max_{r \in \mathcal{R}} \cos(\mathbf{e}_s, \mathbf{e}_r)$$

- First term: **relevance** to cluster centroid $\boldsymbol{\mu}_c$
- Second term: **diversity** from already-selected $\mathcal{R}$
- $\lambda = 0.7$: slightly favours relevance over diversity
- Iterate until $|\mathcal{R}| = K_e = 16$

Effect

Exemplars cover the **full semantic spread** of the cluster, not just the centroid cloud

Centroid vs. MMR Selection



MMR (green) covers the cluster periphery;
centroid-nearest (red) is redundant.

MMR (green) covers the

Two Naming Backends: MLM and LLM

MLM Backend (BERT)

Template:

In this sentence, {ent} is a [MASK].

Sentence: {text}.

Average vocab distributions over K_e exemplars:

$$\bar{p}_v = \frac{1}{K_e} \sum_r p_v^{(r)}$$

Post-filter: remove subwords, stopwords, punctuation.

Output: single token

conference, band, protein

Cannot express compound types:

musical artist, literary genre

LLM Backend (Llama-3 via Ollama)

Two-turn prompt:

Sys: A virtual assistant answers questions...

User: Given entities $\{e_1, \dots, e_{16}\}$, all of the same type, predict the entity type. Respond only with the type name.

Temperature = 0 (deterministic).

Post-process: ≤ 3 words, lowercase, strip fillers.

Output: multi-word labels

political party, research institute

Immune to MLM hallucination artefact

	Multi-word output	No hallucination	Avg BERTScore-F1
MLM (MMR)	✗	✗	0.896
LLM (MMR)	✓	✓	0.956

Source training data:

- **CoNLL-2003**: English newswire, 4 types, 14k sentences
- **Pile-NER**: diverse web corpus, ~ 13 k types, 50k sentences

Target evaluation (CrossNER):

Domain	#Test	#GT Types
AI	431	14
Literature	416	12
Music	465	13
Politics	650	9
Science	543	17

Gold labels used **only for evaluation** — never seen during training or clustering.

Key hyperparameters:

Component	Setting
Detector backbone	BERT-base-cased
Detector LR	1×10^{-5} , 50 epochs
EMA decay	$\alpha = 0.995$
Typing backbone	BERT-base-uncased
Full FT LR	2×10^{-5} , 30 epochs
Soft prompt N_p	5 tokens
Stage 1 LR	3×10^{-3} , 10 epochs
Stage 2 LR	1×10^{-5} , 20 epochs
Triplet margin	$\gamma = 1.0$
Clustering range	$k \in [2, 30]$, step 2
k-means restarts	10 per k
MMR exemplars	$K_e = 16$, $\lambda = 0.7$
LLM	Llama-3 via Ollama

Evaluation Metrics — One per Pipeline Stage

Stage 1 — Detection

Span F1 (exact match)

Precision, Recall, F1 at the span level on test set.

Measures whether DTrans transfers better than BIO.

Stage 2 — Typing

AMI (Adjusted Mutual Information)

Mapping-free: no cluster-to-type assignment.

Robust to permutations, over-seg., over-merging.

0 = random, 1 = perfect.

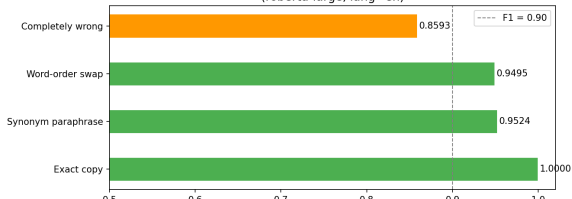
Stage 3 — Naming

BERTScore-F1 (RoBERTa-large)

Semantic similarity of predicted vs. gold type name.

Partial credit for semantically close labels.

BERTScore F1 per test case
(roberta-large, lang=en)



BERTScore interpretation:

Meaningful range is **0.86–1.0**
(even a wrong label scores ≈ 0.86 due to shared embedding space).

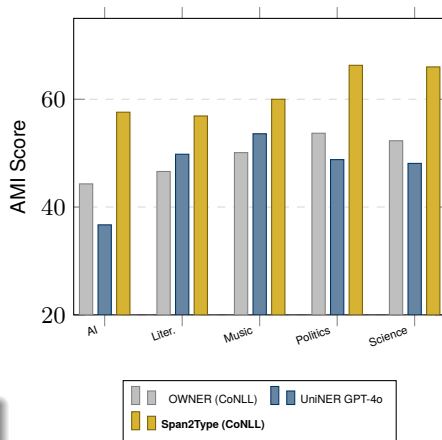
Exact copy = 1.0

Synonym paraphrase ≈ 0.95

Related concept ≈ 0.90

Main Results: End-to-End AMI on CrossNER

Method	AI	Lit	Mus	Pol	Sci	Avg
<i>Unsupervised Baselines</i>						
ChatIE (GPT-4o)	21.8	37.5	43.7	37.2	35.0	35.0
UniNER	33.5	42.8	48.1	40.3	43.7	41.7
UniNER (GPT-4o)	36.7	49.8	53.6	48.8	48.1	47.4
OWNER (CoNLL)	44.3	46.6	50.1	53.7	52.3	49.4
OWNER (Pile)	39.4	49.5	52.5	48.5	50.9	48.2
<i>Proposed</i>						
Span2Type (Pile)	50.0	52.8	56.2	55.8	62.3	55.4
Span2Type (CoNLL)	57.6	56.9	60.0	66.3	66.0	61.4



Key numbers

vs. OWNER: **+12.0 pts** avg AMI

vs. best LLM baseline (UniNER GPT-4o): **+14.0 pts**

Entity Detection F1: Multi-View vs. Single-View

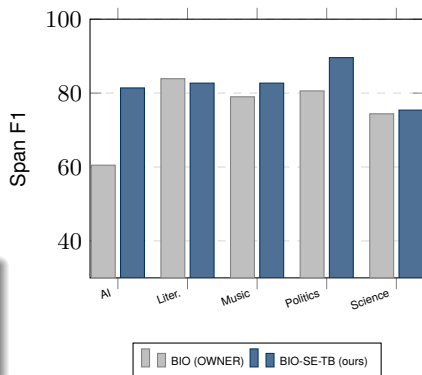
Model	AI	Lit	Mus	Pol	Sci	Avg
PURE	39.8	37.1	33.8	32.4	35.7	35.8
SpanProto	54.1	62.9	59.6	68.7	59.7	61.0
WL-Coref	57.4	68.3	66.9	72.1	63.4	65.6
BIO (OWNER)	60.5	83.9	79.0	80.6	74.4	75.7
BIO-SE-TB	81.4	82.7	82.7	89.6	75.4	82.4

Average gain

OWNER BIO: 75.68

DTrans BIO-SE-TB: **82.35**

Improvement: **+6.67 F1 points**



Literature

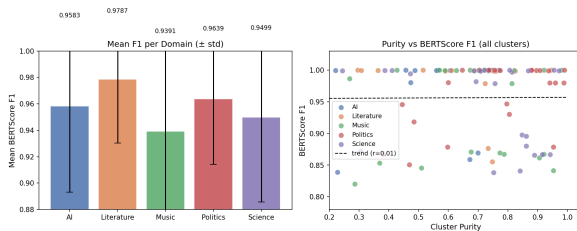
regression: -1.22 — literary entities
(character names, book titles) follow predictable
proper-noun patterns; SE/TB add noise, not signal.

Cluster Naming: BERTScore-F1 Results

Method	AI	Lit	Mus	Pol	Sci	Avg
#Clusters	14	14	22	24	22	—
#GT Types	14	12	13	9	17	—
OWNER MLM	.989	.974	.958	.964	.945	.966
OWNER LLM	.929	.954	.957	.953	.926	.944
S2T Centroid	.904	.887	.895	.883	.894	.892
S2T MLM+MMR	.900	.889	.911	.887	.893	.896
S2T LLM+MMR	.958	.979	.939	.964	.950	.956

Budget-controlled comparison ($n = 16$, same LLM):

- Span2Type-LLM (MMR): 0.956
- OWNER (LLM random): 0.944
- **MMR gain: +0.012 avg**



Left: mean BERTScore-F1 $\pm$ std per domain.

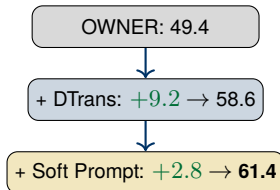
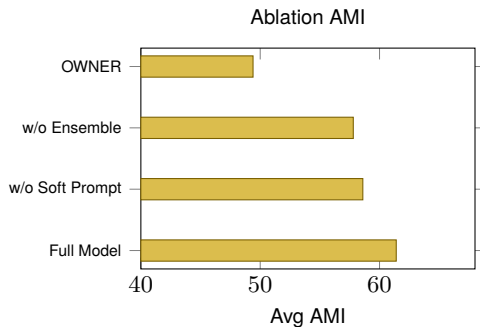
Right: cluster purity vs. BERTScore-F1;
 $r = 0.01$ — naming quality is **independent**
of cluster purity. LLM recovers semantics
even from impure clusters.

Ablation Study: Contribution of Each Component

Model	AI	Lit	Mus	Pol	Sci	Avg
OWNER (baseline)	44.3	46.6	50.1	53.7	52.3	49.4
w/o Soft Prompt	52.3	56.3	57.5	63.9	62.8	58.6
w/o Ensemble MD	49.3	53.0	58.8	64.1	63.8	57.8
Full Model	57.6	56.9	60.0	66.3	66.0	61.4

Conclusions:

- 1 DTrans multi-view: **+9.2** — largest single gain
- 2 Soft prompt: **+2.8** — on top of DTrans
- 3 Both contribute independently: removing either causes a drop



Why CoNLL outperforms Pile-NER?

- CrossNER is Wikipedia-style formal text
- CoNLL-2003 newswire $\approx$ similar syntax and capitalization conventions
- Pile-NER: broader vocab but *noisier, colloquial*
- **Source-target stylistic alignment** > **vocabulary breadth**

Cascade effect: detection $\rightarrow$ AMI

- Span F1 gain: +6.67
- AMI gain: +12.0
- End-to-end gain **exceeds** raw detection gain
- Hypothesis: better spans $\rightarrow$ cleaner [MASK] representations $\rightarrow$ more coherent clusters

When does MMR hurt?

- Music domain: OWNER random (0.957) > MMR (0.939)
- Music clusters are **tight & homogeneous**: *band, genre, album* — unambiguous
- MMR's diversity pressure selects peripheral outliers $\rightarrow$ noise
- **Rule: use MMR for heterogeneous clusters; centroid-nearest for tight clusters**

Naming $\perp$ clustering quality

- Pearson $r = 0.01$ between cluster purity and BERTScore
- LLM can identify dominant type even from impure clusters
- Improving k estimation benefits AMI, not naming quality

1 Multi-view entity detection

DTrans: BIO + SE + TB + Mean Teaching

Span F1: 75.68 $\rightarrow$ **82.35** (+6.67)

Largest gain in AI domain: +20.85

2 Two-stage soft-prompt entity typing

Only 3,840 trainable parameters

Stage 1: prompt warmup (3×10^{-3})

Stage 2: full model (1×10^{-5})

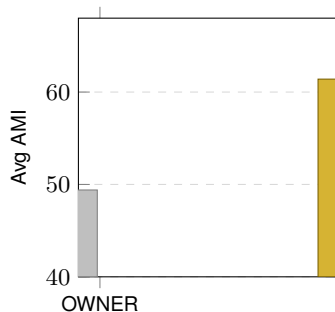
AMI contribution: **+2.8 points**

3 MMR-based cluster naming

$K_e = 16$ diverse exemplars, $\lambda = 0.7$

LLM backend: 0.956 avg BERTScore-F1

+0.012 over random selection (budget-controlled)



End-to-End Result

Span2Type (CoNLL)

61.4 avg AMI

vs. OWNER 49.4

$\Delta = +12.0$ points

Current Limitations:

- Detector still requires labeled **source domain** (newswire → clinical text: large gap)
- **BIC overestimates** k for imbalanced domains (Politics: 24 predicted vs. 9 gold types)
- MLM names: limited to single token
LLM names: requires locally hosted model
- MMR counterproductive for tight, homogeneous clusters (Music)

Future Directions:

- 1 **Domain-adaptive pretraining** or self-training to bridge larger source-target gaps
- 2 **LoRA / adapters** for entity typing — more expressive than 5×768 soft tokens
- 3 **Hierarchical clustering** to produce entity type *ontologies* rather than flat partitions
- 4 **MMR at training time** — diverse in-context retrieval during entity typing
- 5 Evaluate on **biomedical & legal domains** where open-world NER is most needed

Thank You

Questions & Discussion Welcome

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Span2Type: Open World Named Entity Recognition with Automatic Entity Naming

Appendix A — Naming Examples (Qualitative)

Domain	GT Type	LLM Name	GT Type	MLM Name
AI	researcher	computer scientist	conference	conference
Literature	writer	novelist	book	novel
Music	musicalartist	musician	band	band
Politics	politician	political leader	politicalparty	democrat
Science	protein	biological protein	scientist	physicist

Failure patterns:

- **Specificity mismatch:** LLM predicts hyponym (*actor* for *person*)
- **Compound label penalty:** `musicalartist` vs. *musical artist* — semantically correct but tokenization mismatch reduces BERTScore
- **MLM hallucination:** *vegetarian* — BERT defaults to high-frequency unrelated token for OOD context. LLM backend is immune.

Appendix B — Literature Domain Per-Cluster Naming

Cluster	GT Type	LLM Name	BERTScore	
C1	person	author	0.994	✓
C2	book	novel	0.998	✓
C3	location	theatre	0.979	~
C4	writer	poet	0.992	✓
C5	book	mythology	0.996	✓
C6	literarygenre	literary genre	0.993	✓
C7	award	literary award	0.998	✓
C8	event	film festival	0.877	✗
C13	literarygenre	literary work	0.855	✗

Literature = highest domain mean BERTScore-F1 = **0.979**

11 of 14 clusters score ≥ 0.99 (green).

C8 failure: specificity mismatch (*film festival* vs. broad event).

C13 failure: compound string `literarygenre` penalizes correct multi-word name.

LLM-prompting baselines (UniNER, ChatIE) ask the LLM to identify AND type entities in one pass.

Three structural weaknesses vs. Span2Type:

- ❶ **No explicit span detector:** zero-shot extraction limits recall; the LLM's span boundary decisions are not trained discriminatively on entity data.
- ❷ **Inconsistent type strings:** open-vocabulary output from the LLM produces “*a person working in politics*” and “*politician*” for the same type — clustering these together is impossible and AMI scores drop.
- ❸ **Structural mismatch with AMI:** AMI measures agreement between predicted and gold *span partitions*; LLM-extracted spans differ structurally from gold annotation boundaries, creating systematic offset even when semantics are correct.

Span2Type separates span detection (discriminative), embedding (template), and clustering (unsupervised) into **well-defined, optimized stages** — each addressing one specific failure mode.

Hyperparameter	Value
Backbone	BERT-base-cased (12L, H=768, A=12)
Learning rate	1×10^{-5}
Batch size	32
Training epochs	50
Warmup steps	500
EMA decay α	0.995
Consistency weight λ_w	100
Max sequence length	128 tokens
Confidence threshold δ	0.5 (default)
Source corpus	CoNLL-2003 / Pile-NER
Hardware	2× GPU